

# Demystifying Hyperledger Fabric Framework for Distributed Ledgers and Approach to Evaluate Its Performance

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**Abstract**— Hyperledger Fabric is a highly advanced blockchain framework for enterprise level organizations. Hyperledger Fabric is designed to establish either permissioned or private blockchain networks. This means that only endorsed members and parties are able to join the network, and ledgers are only accessible to those parties that have been granted permission to access them. Such environment is ideal for organizations that needs a high degree of privacy and security in their ledgers. Another key feature of Fabric is its endorsement policy, which permits to define flexible and customizable transaction validation rules. As a result, organizations can define which members are required to sign off on transactions before they can be added to the blockchain. The Fabric framework also includes a numeral other features that make it suitable for enterprise-level use such as support for multiple consensus algorithms, modular architecture and private data collections that permits integration of external services. Its modular nature, support for several other consensus algorithms and customizable endorsement policy make it a suitable for businesses looking to setup their own blockchain networks. This research work is disclosing the practical approach to implement Hyperledger Fabric along with the approach to evaluate its performance primarily throughput and latency.

**Index Terms**—Hyperledger Fabric, Hyperledger Caliper, Blockchain, Fabric Performance, Fabric Latency

## I. INTRODUCTION

Hyperledger Fabric is one of the open source and enterprise-grade frameworks that permits businesses and organizations to design and deploy their own blockchain networks [1][2]. Hyperledger framework works under the Linux Foundation and is designed to stipulate a modular architecture that can be tailored to meet the certain needs of distinctive enterprise applications. Hyperledger Fabric facilitate the establishment of permissioned blockchain networks, which means that only authorized members are allowed to join and interact with the network [3]. Thus, it is ideal for enterprise applications that demands a high degree of privacy, security, and scalability. The Fabric framework also includes a numeral other features that make it suitable for enterprise-level use such as support for multiple consensus algorithms, modular architecture and

private data collections that permits integration of external services.

Hyperledger Fabric is very much flexible and modular and permit organizations to tailor their blockchain network to meet their certain requirements. It provides a rich support for numerous consensus algorithms, including Raft and Practical Byzantine Fault Tolerance (PBFT) which can be adopted based on the network's performance requirements. Moreover, Fabric supports the concept of smart contracts also popularly known as "chaincode," and can be programmed in various programming languages such as Go or Node.js.

Another key feature of Fabric is its endorsement policy, which permits to define flexible and customizable transaction validation rules. As a result, organizations can define which members are required to sign off on transactions before they can be added to the blockchain. In nutshell, Fabric is an extremely customizable and potent blockchain framework that is appropriate for a comprehensive range of enterprise applications. Its modular nature, support for several other consensus algorithms and customizable endorsement policy make it a suitable for businesses looking to setup their own blockchain networks [4].

Every technology has pros and cons which must be addressed wisely. Though, Fabric is a powerful and flexible framework that is well-suited for enterprise-level businesses and organizations, yet it is suffering from some issues that need to be addressed. Firstly, Hyperledger Fabric is a complex framework that requires a substantial amount of technical proficiency to set up and manage. This can make it very challenging for businesses with inadequate technical resources to adopt and implement [5-7]. Another challenge is related to scalability. Though Hyperledger Fabric has made significant improvements in its scalability over the years, yet it is facing several challenges while handling high transaction volumes. This is one of the biggest barrier in its adoption for businesses with large-scale use cases. Another challenge is, integrating Hyperledger Fabric with existing enterprise systems, especially when dealing with legacy systems or proprietary technologies.

Hyperledger Fabric is aimed for enterprise level businesses, thus it requires a well-defined governance model to guarantee that all participants on the network are following the rules and regulations. Moreover, presently there is a shortage of blockchain experts, thus making it very challenging for businesses to find the talent they need to productively implement Hyperledger Fabric. Thus, it is clear that even though Hyperledger Fabric is a powerful tool but primarily utilized by larger organizations due to its complexity and lack of expertise. This study is an attempt to disclose its technicalities in a simpler way, approach to implement Hyperledger Fabric server and evaluating system performance primarily throughput and latency.

## II. RELATED WORK

Hyperledger Fabric is a blockchain framework which can be used to implement immutable and auditable distributed ledger for several key areas like supply chain management, electronic health records, insurance sector, asset administration, Internet of Things (IoT) applications and many more. There are several blockchain frameworks like Ethereum, Hyperledger Sawtooth and R3 Corda but among all these Hyperledger Fabric is the most effective due to its low latency and high throughput. Table 1 is depicting all those studies which have performed empirical studies and evaluated performances of multiple blockchain frameworks.

Table 1. Published comparative studies on Blockchain framework

| Citation | Comparison                                  | Year | Outcome                                                |
|----------|---------------------------------------------|------|--------------------------------------------------------|
| [8]      | Ethereum with Hyperledger Fabric            | 2017 | Fabric outperformed                                    |
| [9]      | Ethereum with Hyperledger Fabric            | 2018 | Fabric outperformed                                    |
| [10]     | Ethereum with Hyperledger Fabric            | 2020 | Fabric outperformed                                    |
| [11]     | Ethereum, Fabric, Quaram and R3 Corda       | 2020 | Corda outperformed in latency and fabric in throughput |
| [12]     | Hyperledger Sawtooth and Hyperledger fabric | 2021 | Fabric outperformed                                    |
| [13]     | Hyperledger Iroha, Sawtooth and Fabric      | 2022 | Fabric outperformed                                    |
| [14]     | Ethereum with Hyperledger Fabric            | 2022 | Fabric outperformed                                    |
| [15]     | Ethereum, Parity and Hyperledger Fabric     | 2022 | Fabric outperformed                                    |

The related work indicates that in majority of the cases Hyperledger Fabric performed well with respect to throughput and latency. This framework is suitable for applications areas such as finance and banking, healthcare, identity and access management, real-estate, energy and insurance sector. A comprehensive analysis of sector wise adoption of blockchain is illustrated in the Figure 1. It is clear from this figure that blockchain is widely adopted for Internet of Things (IoT) and energy sector followed by healthcare and supply chain

management system. Artificial intelligence is another area where Blockchain can play a vital role [16][17].

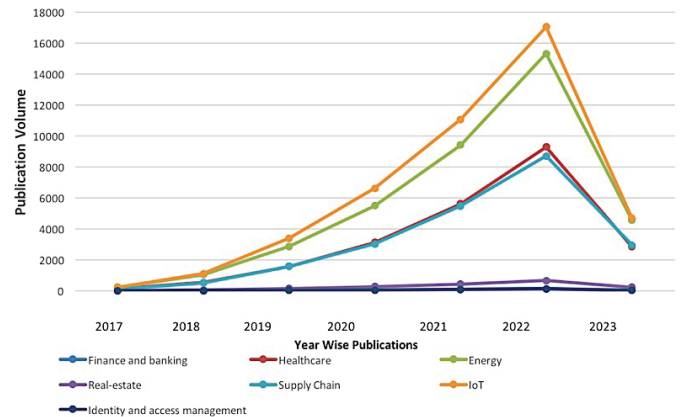


Fig. 1. Sector wise adoption of Blockchain

All these sectors are very sensitive in nature and require a blockchain environment which can deliver best performance in context to throughput and latency. Thus, blockchain framework selection is crucial and several studies has proved that hyperledger fabric framework can delivery high throughput and low latency as compared to others. These studies are mentioned in the Table 1. But due to its complexity and lack of expertise, this framework is less adopted and needs attention.

## III. OBJECTIVES

This study is aimed to demystify the technicalities associated with the Hyperledger Fabric and approach to implement Fabric server. Another objective is to disclose tools and techniques for evaluating system performance primarily throughput and latency by designing a test network and injecting workload (transactions) through fabric tools such as Hyperledger Caliper.

## IV. METHODOLOGY

The proposed approach is to achieve the objectives is to develop a test network in Hyperledger Fabric framework and then evaluate its performance using the Hyperledger Caliper.

Table 2. Test-Network Components

| Component             | Details                                                                                   |
|-----------------------|-------------------------------------------------------------------------------------------|
| Organizations         | Can be visualized as profit or not-profit entities running their business.                |
| Peers                 | Can be visualized as employees/staff/students or any other stake holders.                 |
| Orderer               | A component responsible for generating blocks.                                            |
| Channel               | All above components must join a common channel to send and received transactions/blocks. |
| Certificate Authority | A fabric component responsible to issue certificates to join the blockchain network       |

A test network is setup with the components listed in the Table 2. This test network depicts an environment where there are two organizations with one peer each, a single orderer, certificate authority for every organization and a single

common channel. This test network is also depicted in the Figure 2.

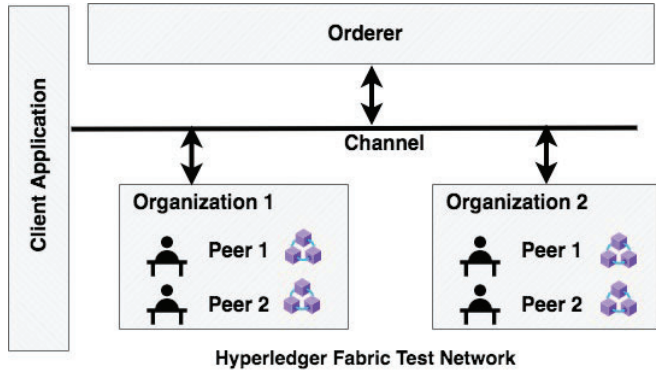


Fig. 2. Hyperledger fabric test network

## V. EXPERIMENT

The test network requires installation of several utilities on the Linux machine. The details of these utilities are illustrated in the Table 3.

Table 3. Prerequisites for the test network

| Software           | Version  | Purpose                                        |
|--------------------|----------|------------------------------------------------|
| Docker             | 20.10.12 | Packaging application into container           |
| cURL               | 7.71.1   | To make http requests                          |
| Ubuntu             | 20.04    | Underlying operating system                    |
| NPM                | 7.24.0   | Node package manger                            |
| Node.js            | 16.10.0  | Node.js and Hyperledger fabric SDK for Node.js |
| Hyperledger Fabric | 2.3.1    | Blockchain Framework                           |
| Fabric Ca          | 1.4.9    | Client Certificate Authority                   |
| Fabric Peer        | 2.3.1    | Peer denotes a client                          |
| Peer Orderer       | 2.3.1    | Orderer organization                           |
| Chaincode          | 2.3.1    | Smart Contract                                 |

This study used Node.js programming language for chaincode to write the business logic and to retrieve and write transactions on the ledger. Bringing up this test network involves a predefined sequence which is listed below.

- Invoking certificate authorities.
- Generating crypto-material for organizations including orderer and peers.
- Bringing up remaining network components configured with docker-compose.
- Bringing channel up requires generating configuration transaction, genesis block with ordering service and joining all peers with this channel.
- Packaging chaincode, installing it on peers, approving chaincode definition for organizations and committing chanincode.

The above steps are automatically followed when a

network.sh script is executed which is bundled with fabric binaries. At any time, new peers can be added to the existing organizations. Adding new peers onto the network require following steps.

- Create crypto material for the new peer.
- Then create a docker-compose file containing this peer.
- Make this peer join the existing channel
- Install the existing chaincode onto the newly created peer.

Thereafter hyperledger caliper is installed and assigned the workload to throw transactions onto this newly created blockchain test network. Hyperledger caliper was configured to inject hundreds of transactions at different rates to evaluate the performance of the network in context of latency and throughput. The experiment was conducted twice, once by injecting transactions at 100 transactions per second (TPS) and then at 200 TPS. A total of 1100 transactions were injected both the times.

## VI. RESULT ANALYSIS

The Hyperledger Calliper, a benchmarking tool for blockchain performance, is used to test the performance of the suggested paradigm. It can be set up to produce a lot of workload on the Fabric. Throughput and latency are employed as metrics to gauge performance. The proposed model will be used on Hyperledger Fabric because of the platform's increased industry popularity as proximately 400 production applications and proofs of concept are currently using it. In order to execute the experiment, two organizations with two peers each were created.

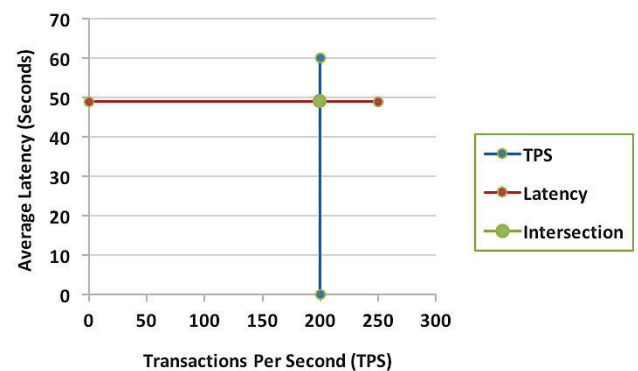


Fig. 3. Average latency with 100 TPS

During experiment, a total of 1100 transactions were injected at 100 TPS. Figure 3 illustrates that system took 49 seconds to commit all the transactions. In the second round, the transaction rate was changed to 200 TPS and this time the system took 53 seconds to commit all the transactions. The results are illustrated in the Figure 4.

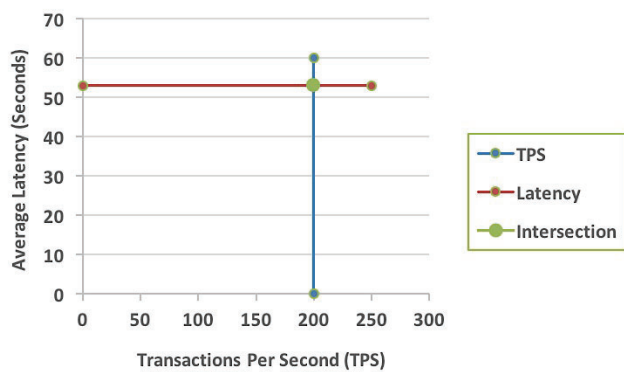


Fig. 4. Average latency with 200 TPS

This study also computed the throughput by first injecting transactions at 100 TPS and then 200 TPS. The results are illustrated in the Figure 5 and Figure 6 respectively.

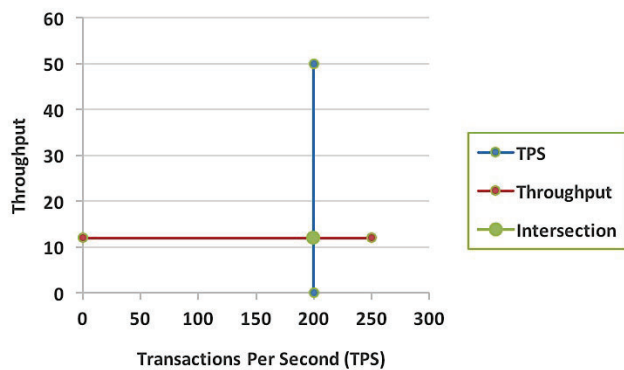


Fig. 5. Average throughput with 100 TPS

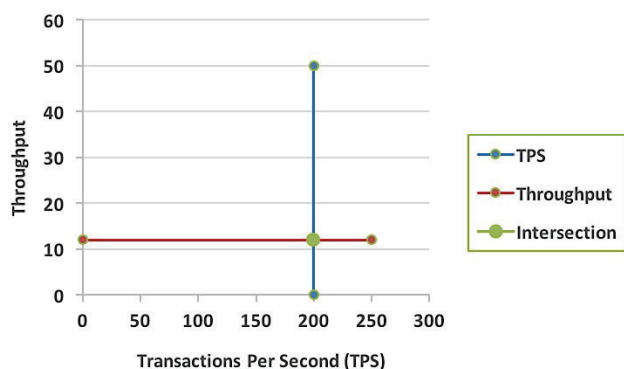


Fig. 6. Average throughput with 200 TPS

It is observed that throughput is 12 TPS regardless of varying transaction rate, 200 TPS and 100 TPS respectively.

## VII. CONCLUSION

This study is demystifying the way to install and use hyperledger framework which is one of the most effective frameworks of blockchain technology. This study also demonstrates the way to setup a test network which simulates a

multi-organization scenario having multiple peers interacting with the shared ledger. This study not even setup the test network but also exhibits the approach to evaluate the network performance. This study proves that hyperledger caliper can be configured to throw huge workload on the network that too with different transaction rates and network can be closely monitored to calculate the overall throughput and latency.

In order to execute the experimental study, two organisations with two peers each were created. These organisations are likely to be profit or non-profit institutions or companies. All peers and organizations were part of a common channel and have similar chaincode.

Only latency and throughput were used as the metrics for testing the suggested model. A total of 1100 transactions were introduced into the Hyperledger network at different rates of 100 and 200 TPS (transactions per second) in order to measure the throughput and latency. This study demonstrates that the throughput remains unaffected even after varying the transaction rates but shift in transaction rates had a slight effect on the latency.

The only flaw in this study is that it does not track throughput and latency as organizations and peers multiply. This is known as scalability, and it will be important to assess how scalability affects network performance in upcoming research. This study suggests that the network might be tested with many peers and many organizations to determine how scalability affects latency and throughput. Testing the hyperledger network when transactions originate from a remote place is another unexplored field of research. Given the need for actual nodes and cloud services, this would be more difficult.

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